A Low-Frequency Momentum Trading Strategy

Harris & Yilmaz (2008), and an Application to BIST100

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Section 4 · Trend Following

Motivation

The paper investigates whether momentum signals should be extracted from the raw exchange rate or from its low-frequency trend component.

Main idea:

$$S_t = S_t^* + v_t$$

where

- S_t : observed exchange rate
- S_t^* : low-frequency trend component
- v_t : high-frequency noise component

Hypothesis:

Momentum in S_t^*

contains more useful information than momentum in the raw exchange rate.

Traditional Momentum: Moving Average Rules

The benchmark strategy uses moving average (MA) rules.

$$MA(m,n) = \frac{1}{m}\sum_{i=0}^{m-1} S_{t-i} - \frac{1}{n}\sum_{i=0}^{n-1} S_{t-i}$$

Trading signals:

$$MA(m,n) > 0 \Rightarrow Buy$$

$$MA(m,n) < 0 \Rightarrow Sell$$

Benchmark rules tested:

$$MA(1,2), MA(1,3), MA(1,4), MA(1,5),$$

$$MA(1,6), MA(2,3), MA(2,6), MA(3,6)$$

Limitations of Moving Average Rules

Moving average rules operate directly on the observed exchange rate.

$$S_t = S_t^* + v_t$$

As a result, trading signals are affected by both:

- Persistent trend information
- Short-term noise

Trade-off:

- Short MA: responsive but noisy
- Long MA: smoother but slower

The paper proposes separating trend from noise before generating momentum signals.

Model 1: Hodrick-Prescott Filter

Purpose:

Estimate the low-frequency trend component of the exchange rate.

HP filter solves:

$$\min_{\{S_t^*\}} \sum_{t=1}^{T} (S_t - S_t^*)^2 + \lambda \sum_{t=2}^{T-1} \left[(S_{t+1}^* - S_t^*) - (S_t^* - S_{t-1}^*) \right]^2$$

Interpretation:

- First term: fit the observed exchange rate
- Second term: penalize excessive trend fluctuations
- λ: smoothing parameter

Baseline value:

$$\lambda = 14,400$$

Model 2: Kernel Regression

Purpose:
Estimate the local nonlinear trend of the exchange rate.

$$S_t^* = \frac{\sum_\tau K_h(t-\tau) S_\tau}{\sum_\tau K_h(t-\tau)}$$

Characteristics:

- Nearby observations receive larger weights
- Distant observations receive smaller weights
- Bandwidth h controls smoothness

Kernel specifications:

- Gaussian
- Parabolic (Epanechnikov)
- Triangular
- Cosine

Kernel Types: Intuition

Gaussian

- Smoothest weighting scheme
- Uses all observations
- Slowest reaction to turning points

Triangular

- Linear decay in weights
- Reacts faster to trend changes

Parabolic

- Strong emphasis on nearby observations
- Good balance between smoothing and responsiveness

Cosine

- Similar to Parabolic
- Smooth transition of weights

Low-Frequency Momentum Strategy

Step 1:
Estimate the trend component

$$S_t^*$$

using either:

- HP Filter
- Kernel Regression

Step 2:
Apply momentum to the estimated trend.
Trading signals:

$$S_t^* - S_{t-1}^* > 0 \Rightarrow \textit{Buy}$$

$$S_t^* - S_{t-1}^* < 0 \Rightarrow \textit{Sell}$$

Data

Currencies:

$$EUR,\ JPY,\ GBP,\ CHF,\ SEK,\ NOK, AUD,\ CAD,\ NZD$$

Equally Weighted Portfolio (EWP):

$$EWP = \frac{1}{9}\sum_{i=1}^{9} Currency_i$$

Sample:

- Full sample: June 1986 – May 2008
- Initial estimation: 1986 – 1993
- Out-of-sample evaluation: 1993 – 2008

Data Sources:

- Reuters
- Datastream

Methodology

Recursive estimation procedure:

1. Estimate trend using historical information
2. Roll estimation window forward one month
3. Re-estimate trend
4. Generate buy/sell signal
5. Record out-of-sample performance

Sub-period analysis:

$$1993 - 1998$$

$$1998 - 2003$$

$$2003 - 2008$$

Performance Measures

The paper evaluates strategies using:

1. Directional Accuracy
2. Annualized Return
3. Sharpe Ratio
4. Maximum Drawdown

Sharpe Ratio:

$$Sharpe = \frac{E(R_p) - R_f}{\sigma_p}$$

Maximum Drawdown:

$$MDD = \max\left(\frac{Peak - Trough}{Peak}\right)$$

Carry returns are included.

Transaction costs are ignored.

Moving Average Results

Table 2 Trading Strategy Peformance (Moving Average)

Panel A: Directional Accuracy

	EUR	JPY	GBP	CHF	SEK	NOK	AUD	CAD	NZD	EWP
MA(1, 2)	54.4%	56.7%	41.7%	51.7%	53.3%	50.0%	52.2%	47.8%	51.1%	52.8%
MA(1, 3)	51.7%	57.8%	40.6%	46.1%	53.9%	47.2%	46.7%	48.3%	51.1%	51.7%
MA(1, 4)	51.1%	55.6%	41.1%	48.3%	51.7%	46.7%	48.3%	48.9%	56.1%	51.1%
MA(1, 5)	52.8%	53.3%	42.2%	46.7%	56.1%	47.2%	46.7%	50.0%	56.7%	53.3%
MA(1, 6)	51.7%	51.7%	43.3%	45.0%	57.2%	46.7%	48.3%	51.7%	58.3%	51.7%
MA(2, 3)	46.7%	55.0%	44.4%	46.1%	50.0%	51.7%	47.2%	45.0%	50.0%	46.1%
MA(2, 6)	53.3%	53.3%	47.8%	45.0%	59.4%	53.3%	51.7%	50.6%	58.9%	55.6%
MA(3, 6)	51.1%	50.0%	52.8%	48.3%	61.1%	50.6%	53.3%	50.0%	63.3%	55.0%

Panel B: Return

	EUR	JPY	GBP	CHF	SEK	NOK	AUD	CAD	NZD	EWP
MA(1, 2)	6.14%	0.67%	-2.80%	1.54%	2.43%	0.90%	3.32%	0.58%	4.38%	1.91%
MA(1, 3)	6.24%	2.62%	-3.39%	0.12%	2.38%	-0.37%	-0.21%	0.86%	3.74%	1.33%
MA(1, 4)	4.39%	2.55%	-3.26%	1.59%	-0.31%	-0.89%	-0.28%	1.51%	6.57%	1.32%
MA(1, 5)	4.93%	2.48%	-2.43%	1.24%	1.76%	-0.81%	-1.37%	1.26%	5.07%	1.35%
MA(1, 6)	3.63%	2.91%	-2.43%	-0.65%	2.75%	-0.86%	-0.47%	2.49%	5.58%	1.44%
MA(2, 3)	2.50%	4.27%	-2.97%	-1.22%	-1.22%	0.87%	-2.11%	0.41%	0.54%	0.12%
MA(2, 6)	3.41%	4.74%	0.13%	-1.39%	4.30%	2.55%	0.92%	0.90%	5.33%	2.32%
MA(3, 6)	1.53%	-0.34%	1.01%	0.16%	6.07%	1.11%	2.67%	-1.31%	7.94%	2.09%

Panel C: Sharpe Ratio

	EUR	JPY	GBP	CHF	SEK	NOK	AUD	CAD	NZD	EWP
MA(1, 2)	0.71	0.06	-0.39	0.15	0.24	0.14	0.33	0.06	0.44	0.41
MA(1, 3)	0.72	0.23	-0.47	0.01	0.23	-0.06	-0.02	0.09	0.37	0.27
MA(1, 4)	0.50	0.23	-0.46	0.16	-0.03	-0.13	-0.03	0.16	0.66	0.28
MA(1, 5)	0.56	0.22	-0.34	0.12	0.17	-0.12	-0.14	0.13	0.51	0.28
MA(1, 6)	0.41	0.26	-0.34	-0.06	0.27	-0.13	-0.05	0.26	0.56	0.29
MA(2, 3)	0.28	0.38	-0.42	-0.12	-0.12	0.13	-0.21	0.04	0.05	0.02
MA(2, 6)	0.39	0.43	0.02	-0.14	0.43	0.39	0.09	0.09	0.53	0.45
MA(3, 6)	0.17	-0.03	0.14	0.02	0.61	0.17	0.27	-0.14	0.81	0.40

Panel D: Maximum Drawdown

	EUR	JPY	GBP	CHF	SEK	NOK	AUD	CAD	NZD	EWP
MA(1, 2)	18.1%	37.8%	50.8%	32.2%	37.8%	20.7%	17.3%	33.1%	19.4%	9.6%
MA(1, 3)	15.9%	23.6%	51.7%	38.9%	32.1%	20.7%	29.1%	29.1%	24.9%	11.5%
MA(1, 4)	18.9%	25.2%	48.9%	34.4%	41.0%	27.9%	29.7%	23.3%	16.8%	8.2%
MA(1, 5)	18.9%	24.8%	42.8%	18.2%	26.6%	26.6%	47.8%	26.0%	28.2%	10.0%
MA(1, 6)	25.7%	30.9%	45.3%	26.5%	26.8%	27.7%	36.9%	26.1%	29.1%	13.3%
MA(2, 3)	19.1%	22.5%	48.1%	30.5%	44.1%	24.2%	37.9%	25.3%	40.4%	14.2%
MA(2, 6)	23.3%	16.7%	23.2%	44.5%	32.1%	10.7%	26.4%	33.8%	39.7%	16.2%
MA(3, 6)	25.6%	40.3%	25.8%	34.1%	18.8%	23.7%	31.0%	51.8%	32.8%	15.3%

HP Filter Results

Table 3 Trading Strategy Peformance (Hodrick-Prescott Filter)

Panel A: Directional Accuracy

	EUR	JPY	GBP	CHF	SEK	NOK	AUD	CAD	NZD	EWP
11,520	55.6%	53.3%	57.8%	51.1%	60.0%	53.3%	55.0%	53.3%	61.1%	61.1%
12,960	56.7%	53.3%	57.8%	51.1%	58.3%	53.9%	55.0%	53.3%	61.1%	60.0%
14,400	56.1%	53.9%	57.8%	51.1%	58.9%	54.4%	55.0%	53.9%	61.7%	60.6%
15,840	56.1%	53.3%	57.8%	51.7%	58.9%	53.9%	54.4%	55.0%	62.2%	60.0%
17,820	55.6%	53.3%	57.8%	51.7%	59.4%	53.9%	55.0%	55.0%	61.1%	60.6%

Panel B: Return

	EUR	JPY	GBP	CHF	SEK	NOK	AUD	CAD	NZD	EWP
11,520	5.15%	4.35%	2.48%	3.35%	5.57%	1.44%	6.20%	3.09%	8.61%	4.47%
12,960	5.25%	4.35%	2.48%	3.09%	4.65%	1.49%	6.20%	3.09%	8.61%	4.36%
14,400	5.14%	4.37%	2.33%	3.09%	4.73%	1.71%	6.20%	3.64%	9.20%	4.49%
15,840	5.14%	3.97%	2.33%	3.38%	4.46%	1.42%	6.04%	4.25%	9.38%	4.49%
17,820	4.57%	3.97%	2.33%	3.38%	4.85%	1.35%	6.67%	4.37%	7.81%	4.37%

Panel C: Sharpe Ratio

	EUR	JPY	GBP	CHF	SEK	NOK	AUD	CAD	NZD	EWP
11,520	0.59	0.39	0.35	0.33	0.56	0.22	0.63	0.32	0.88	0.76
12,960	0.60	0.39	0.35	0.31	0.46	0.22	0.63	0.32	0.88	0.74
14,400	0.59	0.39	0.32	0.31	0.47	0.26	0.63	0.38	0.94	0.76
15,840	0.59	0.36	0.32	0.34	0.44	0.21	0.61	0.45	0.96	0.77
17,820	0.52	0.36	0.32	0.34	0.48	0.20	0.68	0.46	0.79	0.76

Panel D: Maximum Drawdown

	EUR	JPY	GBP	CHF	SEK	NOK	AUD	CAD	NZD	EWP
11,520	15.6%	32.5%	16.2%	23.7%	22.2%	31.0%	15.5%	32.3%	24.3%	8.7%
12,960	15.6%	32.5%	16.2%	23.7%	32.8%	30.2%	15.5%	32.3%	24.3%	8.1%
14,400	16.5%	32.5%	17.9%	23.7%	31.6%	26.8%	15.5%	32.3%	24.3%	7.7%
15,840	16.5%	32.5%	17.9%	22.0%	37.6%	26.8%	15.5%	32.3%	24.3%	7.5%
17,820	16.5%	32.5%	17.9%	22.0%	34.7%	26.8%	15.5%	28.9%	24.3%	7.1%

HP filtering significantly improves both returns and risk-adjusted performance.

Kernel Regression Results

Table 4 Trading Strategy Peformance (Nadaraya-Watson Kernel Regression)

Panel A: Directional Accuracy

	EUR	JPY	GBP	CHF	SEK	NOK	AUD	CAD	NZD	EWP
Gaussian	56.7%	48.9%	54.4%	51.7%	56.1%	57.2%	53.3%	53.9%	58.3%	57.8%
Parabolic	56.7%	53.9%	53.9%	53.9%	59.4%	52.2%	55.6%	58.3%	60.0%	62.8%
Triangular	58.3%	53.9%	50.6%	53.3%	60.0%	52.8%	56.7%	57.2%	61.7%	62.8%
Cosine	57.2%	53.9%	53.3%	54.4%	60.0%	52.8%	55.6%	57.8%	61.7%	62.8%

Panel B: Return

	EUR	JPY	GBP	CHF	SEK	NOK	AUD	CAD	NZD	EWP
Gaussian	4.20%	0.70%	0.39%	3.16%	4.24%	3.05%	3.70%	4.32%	5.91%	3.30%
Parabolic	5.38%	4.76%	1.61%	4.90%	5.02%	1.49%	6.75%	3.84%	7.86%	4.62%
Triangular	6.55%	4.81%	0.29%	4.44%	6.22%	1.85%	7.26%	3.89%	8.66%	4.89%
Cosine	5.61%	4.76%	1.38%	5.07%	5.36%	1.78%	6.75%	3.78%	8.72%	4.80%

Panel C: Sharpe Ratio

	EUR	JPY	GBP	CHF	SEK	NOK	AUD	CAD	NZD	EWP
Gaussian	0.48	0.06	0.05	0.32	0.42	0.46	0.37	0.45	0.59	0.56
Parabolic	0.62	0.43	0.22	0.49	0.50	0.22	0.69	0.40	0.80	0.80
Triangular	0.76	0.43	0.04	0.44	0.62	0.28	0.74	0.41	0.88	0.87
Cosine	0.64	0.43	0.19	0.51	0.54	0.27	0.69	0.40	0.89	0.83

Panel D: Maximum Drawdown

	EUR	JPY	GBP	CHF	SEK	NOK	AUD	CAD	NZD	EWP
Gaussian	17.9%	49.9%	26.9%	20.4%	27.6%	12.6%	27.4%	16.5%	26.4%	13.2%
Parabolic	18.6%	28.0%	21.1%	19.7%	22.2%	23.4%	16.9%	29.7%	29.2%	9.8%
Triangular	11.6%	28.0%	32.2%	28.5%	16.8%	20.2%	18.0%	35.1%	24.3%	7.9%
Cosine	18.6%	28.0%	22.3%	19.7%	22.2%	19.1%	16.9%	30.6%	24.3%	9.7%

Kernel-based strategies consistently outperform traditional moving averages.

EWP Performance Comparison I

Method	Accuracy (%)	Return (%)	Sharpe	MDD (%)
MA(1,2)	52.8	1.91	0.41	9.6
MA(1,3)	51.7	1.33	0.27	11.5
MA(1,4)	51.1	1.32	0.28	8.2
MA(1,5)	53.3	1.35	0.28	10.0
MA(1,6)	51.7	1.44	0.29	13.3
MA(2,3)	46.1	0.12	0.02	14.2
MA(2,6)	55.6	2.32	0.45	16.2
MA(3,6)	55.0	2.09	0.40	15.3
HP $\lambda = 11520$	61.1	4.47	0.76	8.7
HP $\lambda = 12960$	60.0	4.36	0.74	8.1

Note: Moving Average strategies are generally weaker than HP and Kernel-based methods.

EWP Performance Comparison II

Method	Accuracy (%)	Return (%)	Sharpe	MDD (%)
HP $\lambda = 14400$	60.6	4.49	0.76	7.7
HP $\lambda = 15840$	60.0	4.49	0.77	7.5
HP $\lambda = 17820$	60.6	4.37	0.76	**7.1**
Gaussian Kernel	57.8	3.30	0.56	13.2
Parabolic Kernel	**62.8**	4.62	0.80	9.8
Triangular Kernel	**62.8**	**4.89**	**0.87**	7.9
Cosine Kernel	**62.8**	4.80	0.83	9.7

Finding: Triangular Kernel gives the strongest overall performance, while HP $\lambda = 17820$ has the lowest maximum drawdown.

Robustness Tests

The authors test robustness by varying:

- HP filter smoothing parameter (λ)
- Kernel bandwidth (h)
- Kernel type
- Sample period

Main Finding:

- Results remain stable across parameter choices.
- Low-frequency momentum is less sensitive than MA rules.
- Performance remains relatively stable across subperiods.

Economic Interpretation

Traditional strategy:

$$S_t - S_{t-1}$$

Low-frequency strategy:

$$S_t^* - S_{t-1}^*$$

The paper argues that:

- Momentum exists primarily in the trend component.
- Noise obscures useful information.
- Trend extraction improves signal quality.

From the paper to BIST100

We reproduce the Harris and Yilmaz idea on Turkish equities rather than currencies, inside the course-mandated backtest frame.

- **Same core idea:** estimate a low-frequency trend, trade the sign of its slope.
- **Trend extractors:** we swap the paper's kernel regression for LOWESS, and keep the HP filter as the principled anchor.
- **Universe:** the BIST100 index for tuning, then its constituents.
- **Evaluation:** Phase-1 sweep, Phase-2 portfolio, regime filter, Monte-Carlo significance, and walk-forward out-of-sample.

Course-mandated backtest setup

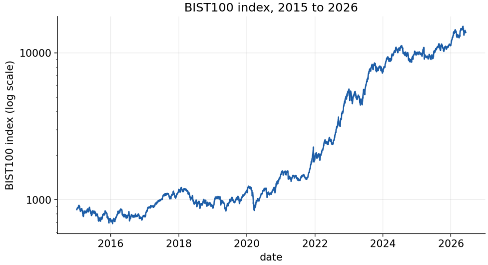
Everything below sits inside a fixed frame that the brief mandates, which we keep unchanged:

- **Engine:** Backtrader **Bars:** daily
- **Initial capital:** 1,000,000 TL
- **Sizing:** `FixedCashSizer`, 100,000 TL notional per order
- **Commission:** 0 **Order type:** market, fills next bar
- **Significance test:** Monte-Carlo Sharpe p-value against random strategies matched on trade frequency, holding period and long/short mix, with $N = 1000$

Two-phase design. In Phase 1 we tune each indicator on the BIST100 index. In Phase 2 we run the winner on the constituents, add a regime filter, aggregate to a portfolio and stress-test it.

The data

- Source: `yfinance`, daily OHLCV
- Window: 2015 to 2026, about 2860 bars
- Phase 1: BIST100 index (`XU100.IS`)
- Phase 2: **84** current constituents
- Survivorship bias acknowledged, not corrected
- `clean.py` rescales single-day jumps above 3×, fixing the 2005 TL redenomination and corporate-action gaps
- Minimum history floor: 750 bars



The Phase-1 training series, on a log scale.

Indicators: four trend extractors

The strategy is trend following. We estimate a smooth trend and trade its slope.

1. **S1, EMA crossover:** $\mathrm{sign}(\mathrm{EMA}_{\mathrm{fast}} - \mathrm{EMA}_{\mathrm{slow}})$
2. **S2, EMA direction:** $\mathrm{sign}\left(\Delta\,\mathrm{EMA}_n\right)$
3. **S3, HP filter:** sign of the slope of the Hodrick–Prescott low-frequency trend, tuning λ $\leftarrow$ **our anchor**
4. **S4, LOWESS:** sign of the slope of a LOWESS-smoothed trend, tuning `frac`

S1 and S2 are the moving-average benchmarks. S3 and S4 are the low-frequency trend strategies: S3 is the HP filter that Harris and Yilmaz anchor on, and S4 is the LOWESS stand-in for their kernel regression, carried as the empirical benchmark.

Pitfall we designed around: look-ahead bias

The brief's bare `hp_filter` and `lowess` solve over the whole series, so the value at time t depends on data after t. That is fine for plots but fatal in a backtest. We wrap them in a causal rolling refit that returns only the right-edge value.

```
def rolling_hp_trend(prices, lam=1600, window=504):
    # at each bar t, refit HP on prices[t-window+1 .. t]
    arr = prices.to_numpy(float); n = len(arr)
    out = np.full(n, np.nan)
    eye = np.eye(window); D = np.diff(eye, n=2, axis=0)
    cho = cho_factor(eye + lam * D.T @ D, lower=True) # pre-factor once
    for t in range(window - 1, n):
        out[t] = cho_solve(cho, arr[t-window+1 : t+1])[-1] # right edge only
    return pd.Series(out, index=prices.index)
```

EMA is causal by construction (`ewm` uses only the past), so only HP and LOWESS need the wrapper.

Signals: sign of the trend slope

Every indicator collapses to a $+1/-1/0$ line, and warmup bars are forced to 0.

```
def hp_direction_signal(prices, lam=1600, window=504):
    trend = rolling_hp_trend(prices, lam=lam, window=window)
    sig = np.sign(trend.diff()).astype(float) # S3: slope sign
    sig.iloc[:window] = 0.0
    return sig.rename("trend")
```

- The signed line is precomputed outside Backtrader and fed in as a data line, so S1 through S4 share one strategy class and only the line differs.
- The `shift(1)` convention follows the brief, where the previous value appears at the current index.

Position sizing & trade logic

No stop-loss or take-profit. The strategy is always in the market and reverses on a sign flip. There is no separate exit rule, matching the brief.

Full-capital deployment. `FixedCashSizer` caps each order at 100k TL, so on a flip we fire $\lfloor \text{equity}/100\text{k} \rfloor$ market orders, about 10 lots for a fresh 1M account. Capital stays fully invested rather than 90% in cash.

```python
def next(self):
    sig = self.trend[0]
    if sig == 0.0: self._prev = sig; return
    if sig > 0 and self._prev <= 0: # flip to long
        if self.position.size < 0: self.close()
        self._deploy(+1) # floor(equity/100k) buys
    elif sig < 0 and self._prev >= 0: # flip to short
        if self.position.size > 0: self.close()
        self._deploy(-1)
    self._prev = sig
```

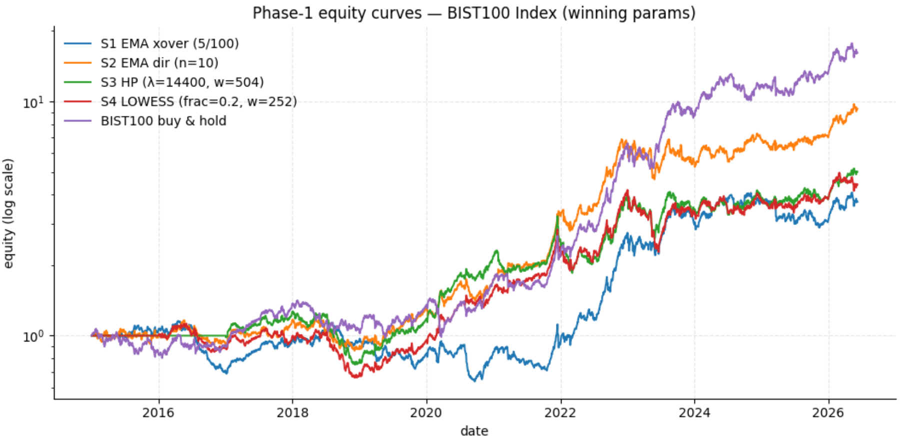
Phase-1 sweep: best config per indicator

Grid search on the BIST100 index, in-sample. Sharpe shown with at least 30 trades.

Strategy	Best params	Sharpe	CAGR	MDD
S4 LOWESS dir.	`frac=0.05, win=252`	**1.11**	25.1%	−25.0%
S3 HP dir.	λ=100, `win=252`	1.04	23.6%	−30.3%
S2 EMA dir.	`n=10`	0.96	21.7%	−32.7%
S1 EMA crossover	5/200	0.70	14.3%	−48.6%

Reference: BIST100 buy-and-hold over the window reaches a Sharpe near 1.1. The slope-of-trend strategies S3 and S4, and EMA direction S2, all beat the classical moving-average crossover S1. That is exactly the Harris and Yilmaz claim.

Phase-1 equity curves



Aggregated equal-weight portfolio

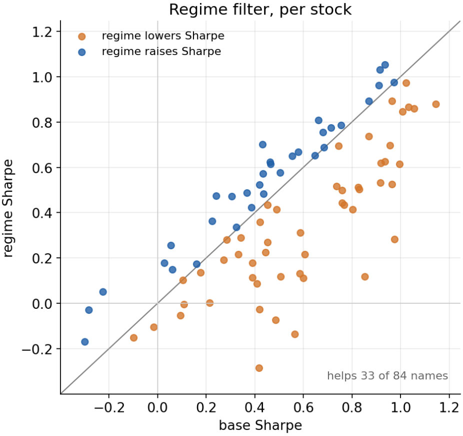
Cross-sectional mean of the per-ticker strategy returns over the 84 names. Diversification roughly triples Sharpe versus the single-name mean.

Variant	Sharpe	CAGR	Ann. vol	MDD	Calmar
Base, no regime	**1.70**	26.2%	14.3%	−17.5%	1.50
BIST100 HP regime	1.36	17.1%	12.2%	−16.9%	1.01

Per-stock mean Sharpe is about 0.60, while the portfolio reaches 1.70. That gap is the cross-sectional diversification benefit. The regime filter lowers return and volatility together.

Idea 1: regime filter

Gate entries by the index trend: go long a stock only when BIST100 sits above its HP value, short only when below. Exits ignore the regime.



Verdict: the regime is a variance reducer, not a Sharpe enhancer. It cuts trade count by about 30% and lowers volatility from 14.3% to 12.2%, but the base portfolio still edges it on Sharpe, 1.70 against 1.36.

The scatter shows why. Points above the diagonal are names the regime helped, below are names it hurt. It rescues the weak left-hand names and clips some of the strong right-hand ones, helping 33 of 84 in net.

Idea 2: LOWESS as an empirical benchmark

We carried S4 LOWESS alongside S3 HP through Phase 2. HP is primary on theory and LOWESS is the empirical challenger.

Portfolio metric	S3 HP	S4 LOWESS
Sharpe (base)	**1.70**	1.68
Sharpe (regime)	1.36	**1.45**
CAGR (base)	26.2%	24.9%
Total return	14.8×	13.1×

On the 2015-start window the two are neck and neck. HP edges LOWESS on the base portfolio, 1.70 against 1.68, and on per-stock significance, 13 against 9 names at $p < 0.01$. LOWESS edges HP only once the regime filter is on. This keeps HP as the primary choice with LOWESS as a close, well-behaved alternative.

Idea 3: conviction sizing & short-side overlay

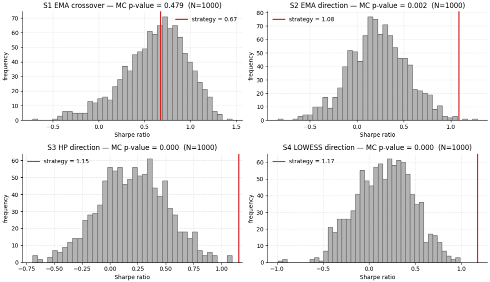
An overlay that is not brief-compliant, reported as a delta against the binary equal-weight baseline. It scales each name's exposure by trend-slope conviction and manages the short side.

Scheme	Sharpe	Δ vs. baseline
binary / both (baseline)	1.67	ref.
conviction (retvol) / both	1.95	+0.28
binary / regime_short	2.24	+0.57
conviction (retvol) / long_only	2.00	+0.33
conviction / regime_short	**2.40**	+0.73

Conviction adds about +0.28 Sharpe, and keeping shorts only when the index regime confirms adds the most. The absolute Sharpes are inflated by the unit-gross daily-rebalance reconstruction, so only the deltas are meaningful and the levers still need re-validation through the real engine.

Significance on the index candidates: Monte-Carlo Sharpe ($N = 1000$)

Strategy	Sharpe	p-value
S4 LOWESS	1.18	**0.000**
S3 HP	1.15	0.000
S2 EMA dir	1.08	0.002
S1 xover	0.67	0.479

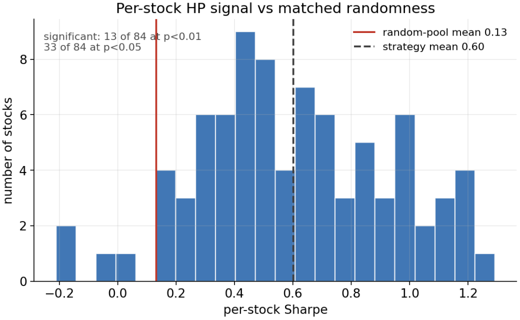


The random pool matches each strategy's trade frequency, holding period and long/short mix. S2, S3 and S4 reject randomness at $p \leq 0.002$. The classical crossover S1 does not ($p = 0.48$): its few long trades are indistinguishable from luck, which is the paper's point.

Significance per stock (S3 HP, 84 constituents)

Each name's HP-direction Sharpe against its own matched-random pool.

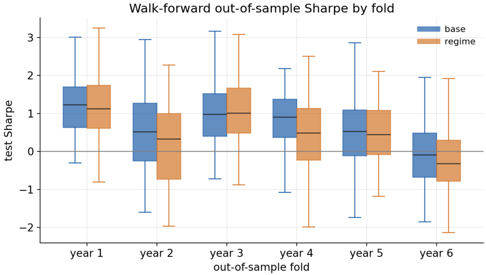
Single-name Sharpes are noisy because always-in-market full-capital shorts on drifting equities produce deep -50% to -90% drawdowns. Still, the mean sits well above the random pool, and the cure for the noise is to aggregate into a portfolio.



Walk-forward out-of-sample (3y train, 1y test)

Re-tune the HP grid on each 3-year window and trade the winner on the next year. This is 397 folds over 68 names, all out-of-sample.

	Base	Regime
Mean test Sharpe	0.68	0.55
% folds > 0	74%	70%
Mean test MDD	−35%	−30%



Out-of-sample Sharpe is positive in about three-quarters of folds. The median stays above zero in every fold except the last. `window=504` wins only once in 397 folds, so we drop it from future sweeps.

Our BIST100 findings

- **Setup:** Backtrader, 1M TL, fixed-cash full deployment, no stop or target, always in market, daily bars, strictly per brief.
- **Indicators:** HP as primary (Harris and Yilmaz) and LOWESS, both wrapped in causal rolling refits to kill look-ahead bias.
- **Signal:** sign of the trend slope, reversing on a flip.
- **Phase 1:** every direction strategy beats the crossover and rejects randomness at $p \leq 0.002$, while the crossover itself does not ($p = 0.48$).
- **Phase 2:** equal-weight portfolio Sharpe of **1.70** for HP and **1.68** for LOWESS, with diversification roughly tripling the single-name Sharpe.
- **Improvements:** the regime filter is a variance reducer, conviction with regime-gated shorts adds the most Sharpe as an overlay, and the walk-forward stays positive out-of-sample.

Conclusion

Main contribution:

$\boxed{\text{Momentum should be measured on the trend component,}}$

$\boxed{\text{not on the raw exchange rate.}}$

Key findings, in the paper and on BIST100:

- Higher directional accuracy
- Higher returns
- Higher Sharpe ratios
- Lower or comparable drawdowns
- Greater robustness

Final takeaway:

$\boxed{\text{The signal is stronger in the filtered trend than in raw prices.}}$